INCH-POUND

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6 March 1992
SUPERSEDING
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30 September 1986

MILITARY SPECIFICATION

# SHEARING MACHINES, METAL SQUARING, MECHANICAL DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers mechanically driven, metal squaring, shearing machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following types. The size shall be determined by material thickness and cutting length capacities as listed in tables I and II. The type and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Gapless Frame

Type II - Gap Frame

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).



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AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, Suite 201, 1500 King Street, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.4 - Shears - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of shearing material in accordance with the requirements specified herein. The machine shall be a self-contained power driven unit and designed to accomplish shearing by the downward movement of a ram mounted upper shear blade (knife) past a stationary lower shear blade. The machine's design and construction shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1) or metric (see 6.5.2), units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials or readouts used to indicate workpiece position shall be graduated in increments not larger than 0.001 inch. All controls shall be permanently and legibly engraved or etched on a nonglare background that can be read from the operator's normal position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the shear gauges for each axis of travel specified. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of travel for each axis on which installed. The system resolution shall be +0.0010 inch. The system accuracy shall be ±0.0010 inch. The repeatability shall be ±0.0005 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate gauge location in inch or metric units and be switchable from inch to metric. An input power source fluctuation of ±10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of gauge travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel.



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The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1.2.2 Electronic display. The electronic display for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle. The unit shall display not less than six digits per axis. The unit shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch-metric conversion, and absolute zero recall. The unit shall be installed so as to be readily accessible and easily read from the operator's work station. The unit shall be mounted to withstand shock and vibration generated by the machine.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, electrical, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B11.4. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.4, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Covers, guards, and other safety devices. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and thickness, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. All guards shall allow access to the guarded part(s).

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a sound level meter at slow response.



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3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, service, transportation, or storage of the machine, and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions in effect on the date of the contract. The operation of the machine shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible. Unless otherwise specified (see 6.2), a centralized manual one-shot lubrication system with positive adjustable lubricant delivery rate, a 24 hour operating capacity, automatic high pressure relief mechanism, and cleanable or replaceable in-line filter shall be supplied and installed on the machine. Means for determining lubricant level in each reservoir shall be by sight gauge, warning light or a similar device.

3.2.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.



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3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. Gears of drive trains shall conform to the provisions of AGMA 390.03a, AGMA 2001-B88, and AGMA 2000-A88 for the inch-pound system of measurement or ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The shearing machine shall consist of not less than the following major components:

3.4.1 Frame. The frame assembly shall support all components of the machine at a level of rigidity required to perform shearing operations specified herein without permanent deformation or failure. The frame shall include two end housings, a workpiece table and an upper crosshead or cross members joined together to form a single unit. The frame members shall be constructed of either ductile iron castings or a welded steel fabrication. The members shall be joined either with a bolted interlocking construction or as a welded assembly. The frame shall support the top of the workpiece table at a working height of not less than 32 inches, and not more than 40 inches from the floor surface.

3.4.2 End housings. End housings shall be either ductile iron castings or steel plate. The type I machine shall be provided with end housings of the gapless or straight design. End housings for the type II machine shall conform to the requirements of table II to provide gaps or workpiece openings for obtaining off-cuts of not less than 18 inches wide from workpieces of any length. The end housings shall be provided with hardened guides for supporting and guiding the upper shear blade to maintain precise blade clearance to accomplish the shearing operations specified herein. A type II machine may be substituted for a type I machine, subject to the approval of the contracting officer (see 6.2). The base of the end housings shall have means to permit leveling and fastening the machine to the floor or foundation. The end housings shall be provided with manual quick adjustment blade clearance levers or manual adjustment devices for obtaining and maintaining clearance between the upper and lower shear knives.



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3.4.3 Workpiece table. The workpiece table or bed shall be constructed of either ductile iron casting or steel plate, with the top surface finished to 250 micro inches or less, and flat within a tolerance of + 0.005 inch per linear foot. The table shall be provided with either a solid bed or a plate and rail bed type top. The table shall securely support the lower shear blade and shall be provided with means for facilitating workpiece handling. The top surface of solid bed type tops shall be provided with slots for attaching workpiece positioning gauges as specified herein. The plate and rail bed type tables shall be constructed to accommodate the attachment of workpiece positioning gauges. When specified (see 6.2), the table top of machines rated 1/4 inch or greater shall be provided with a ball transfer arrangement consisting of steel balls of not less than 3/4 inch diameter. The balls shall be mounted in self-cleaning cups and arranged in a manner suitable for properly supporting full capacity shearing.

3.4.4 Ram. The ram shall be constructed of either a ductile iron casting or a fabricated steel plate and shall have the strength and rigidity required to hold and support the moveable shear knife in the shearing action. The ram shall be guided through the shearing stroke by hardened ram guides arranged to hold the shear knife in alignment or by mounting the ram in pivot bearings in the side frames. The ram guides or pivot bearings shall guide the ram on a vertical shearing plane of not more than two angular degrees off the shearing plane of the stationary blade.

3.4.4.1 Shear blades. Shearing machines rated less than 1/2 inch capacity shall be provided with high carbon, high chrome tool steel shear blades. The shear blades shall be hardened to not less than 57 Rockwell C, and machined for endurance and accuracy. The shear blades of machines rated 3/8 inch and greater shall be supplied with high carbon, high chrome, shock resistant tool steel shear blades. Shear blades shall be reversible, shall have four cutting edges, and shall be hardened and ground to comply with the manufacturer's standard practice. When specified (see 6.2), an extra set of shear blades shall be furnished with the machine.

3.4.5 Drive system. The machine shall have an electric motor driven mechanical drive system capable of initiating and controlling the reciprocating stroke of the ram. A dynamically balanced cast iron or steel fly-wheel shall supply supplemental power to the drive system. An eccentric shaft shall convert the rotary motion of the motor and flywheel into the reciprocating motion required by the ram to complete the shearing stroke. The eccentric shaft shall rotate within anti-friction bearings and shall be directly supported by the end housings of the frame. The energy developed by the drive system shall be controlled by a clutch capable of transmitting the maximum developed torque to the eccentric shaft and ram. The clutch shall be provided with a means for stopping the motion of the ram at the top of the ram stroke when the control is set for single stroke operation, and shall conform to the requirements of ANSI B11.4.

3.4.5.1 Drive system operating controls. The machine shall be provided with controls for operating the ram in both single stroke and continuous (repeating) stroke operation. The drive system shall be operated by a remote control foot switch, and shall have provisions for manually advancing the ram to check blade clearances.



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3.4.6 Holddown system. The machine shall be provided with a holddown system capable of securing any size workpiece up to and including the rated capacity of the machine. The holddown system shall consist of either spring loaded, hydraulic, or pneumatic type plunger units, with a centerline spacing of not more than 12 inches across the full workpiece opening of the machine. The plunger units shall apply uniform holddown pressure, preventing any workpiece movement that would impede the required shearing performance specified herein. The plunger units shall automatically secure the workpiece to the machine table during the ram downstroke and release the workpiece on the return stroke, without causing damage to the workpiece.

3.4.7 Gauges. Unless otherwise specified (see 6.2), the machine shall be provided with a back gauge, front gauge, and side squaring gauges. The gauges shall accommodate workpiece sizes up to the rated capacity of the machine.

3.4.7.1 Back gauge. The back gauge shall be furnished on the rear of the machine to provide an adjustable reference edge between the back edge of the workpiece and the lower shear knife. Unless otherwise specified, the machine shall have either a power operated Acme lead screw back gauge with backlash compensation or a power operated precision ball screw back gauge with built in dampening springs to cushion impact loads. When only one type lead screw is acceptable, the particular type shall be specified (see 6.2). The back gauge position indicator shall display not more than 0.001 inch calibrations in the distance between the back gauge and the lower shear blade. The position indicator shall be located on the machine in a position that is readable from the operator's station. A means of adjustment shall be provided for correcting any back gauge misalignment and adjustments shall be provided for taper cutting. When specified (see 6.2), the machine shall be provided with means for swing up of the back gauge to permit shearing of workpieces whose dimensions exceed the back gauge range.

3.4.7.2 Front gauge. The front gauge shall consist of not less than two heavy duty support arms mounted at 90 degrees to the shearing blade and furnished with slots and recessed precision stainless steel scale graduated in not more than 1/16 inch increments. The support arms shall be equipped with not less than two each disappearing type, adjustable work stops that fit snugly in the accurately machined slots. These gauge mechanism stops shall be of the manufacturer's standard design and shall have the capacity to adjust the workpiece for any position between 6 inches or less to the maximum capacity as shown in table I or II.

3.4.7.3 Side squaring gauges. Both the right and left side of the machine shall be provided with a detachable side squaring gauge to position the edge of the workpiece at right angles to the lower shear blade. Graduated scales calibrated in not more than 1/16 of an inch and adjustable to compensate for blade wear shall be recessed in either the side squaring gauge or the surface of the bed to facilitate workpiece positioning. When specified (see 6.2), an extension side squaring gauge with length as specified shall be provided with the machine. The extension side squaring gauge shall be provided with a recessed adjustable scale graduated in sixteenths of an inch and not less than two adjustable stops.



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3.4.7.4 Light beam shearing gauge. When specified (see 6.2), the machine shall be provided with a fluorescent type light beam shearing gauge that utilizes light beam shadows at the shear line to facilitate shearing to a scribed line on the workpiece.

3.4.7.5 Angle gauge. When specified (see 6.2), the machine shall be provided with an adjustable angle gauge for shearing workpieces at angles without the use of scribed lines.

3.4.7.6 Slitting gauge. When specified (see 6.2), the gap type shearing machine shall be provided with a gauge for guiding the workpiece in slitting operations.

3.4.8 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz(HZ), dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.8.1 Motors. Motors shall meet the requirements of 3.4.8. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.8.2 Solid state components. Except for relays allowed in 3.4.8.1, solid state design shall be used throughout for electronic components.

3.4.8.3 Worklight. When specified (see 6.2), a fluorescent type worklight shall be provided. The light shall have an on-off switch and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.8.4 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I or II for the type and size machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.



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TABLE I. Type I - gapless frame, sizes and capacities.

| Machine size | Workpiece thickness | Workpiece width (feet) | Rake angle (inches per foot) | Shear speed (Strokes per minute - full length at rated rake) | Back gauge capacity (inches) | Front gauge capacity (inches) |
|---|---|---|---|---|---|---|
| 16 ga.x 4 | 16 gauge | 4 | 3/8 | 60 | 24 | 32 |
| 16 ga.x 5 | 16 gauge | 5 | 3/8 | 60 | 24 | 32 |
| 16 ga.x 6 | 16 gauge | 6 | 5/16 | 60 | 24 | 32 |
| 16 ga.x12 | 16 gague | 12 | 1/4 | 60 | 24 | 36 |
| 14 ga.x 3 | 14 gauge | 3 | 13/32 | 60 | 24 | 32 |
| 14 ga.x 4 | 14 gauge | 4 | 13/32 | 60 | 24 | 32 |
| 14 ga.x 5 | 14 gauge | 5 | 13/32 | 60 | 24 | 32 |
| 14 ga.x 6 | 14 gauge | 6 | 13/32 | 60 | 24 | 32 |
| 14 ga.x 8 | 14 gauge | 9 | 3/8 | 60 | 24 | 36 |
| 14 ga.x10 | 14 gauge | 10 | 3/8 | 60 | 24 | 36 |
| 14 ga.x12 | 14 gauge | 12 | 3/8 | 60 | 24 | 36 |
| 12 ga.x 4 | 12 gauge | 4 | 3/8 | 60 | 24 | 36 |
| 12 ga.x 6 | 12 gauge | 6 | 3/8 | 60 | 24 | 36 |
| 12 ga.x 8 | 12 gauge | 8 | 3/8 | 60 | 24 | 36 |
| 12 ga.x10 | 12 gauge | 10 | 3/8 | 60 | 24 | 36 |
| 12 ga.x12 | 12 gauge | 12 | 3/8 | 60 | 24 | 36 |
| 12 ga.x14 | 12 gauge | 14 | 3/16 | 60 | 36 | 50 |
| 10 ga.x 4 | 10 gauge | 4 | 5/16 | 60 | 24 | 24 |
| 10 ga.x 6 | 10 gauge | 6 | 3/8 | 60 | 24 | 36 |
| 10 ga.x 8 | 10 gauge | 8 | 3/8 | 60 | 24 | 36 |
| 10 ga.x10 | 10 gauge | 10 | 3/8 | 60 | 24 | 41 |
| 10 ga.x12 | 10 gauge | 12 | 7/32 | 60 | 24 | 41 |
| 10 ga.x14 | 10 gauge | 14 | 9/64 | 60 | 24 | 41 |
| 10 ga.x16 | 10 gauge | 16 | 15/64 | 60 | 36 | 50 |
| 3/16 x 4 | 3/16 inch | 4 | 7/16 | 60 | 24 | 36 |
| 3/16 x 6 | 3/16 inch | 6 | 3/8 | 60 | 24 | 36 |
| 3/16 x 8 | 3/16 inch | 8 | 3/8 | 60 | 24 | 36 |
| 3/16 x 10 | 3/16 inch | 10 | 3/8 | 60 | 24 | 41 |
| 3/16 x 12 | 3/16 inch | 12 | 3/8 | 55 | 24 | 41 |
| 3/16 x 14 | 3/16 inch | 14 | 15/64 | 55 | 36 | 50 |
| 3/16 x 16 | 3/16 inch | 16 | 7/32 | 50 | 36 | 54 |
| 3/16 x 18 | 3/16 inch | 18 | 7/32 | 50 | 36 | 56 |
| 1/4 x 4 | 1/4 inch | 4 | 3/8 | 60 | 24 | 41 |
| 1/4 x 6 | 1/4 inch | 6 | 3/8 | 60 | 24 | 41 |
| 1/4 x 8 | 1/4 inch | 8 | 3/8 | 60 | 36 | 41 |
| 1/4 x 10 | 1/4 inch | 10 | 3/8 | 55 | 36 | 41 |
| 1/4 x 12 | 1/4 inch | 12 | 3/8 | 55 | 36 | 41 |
| 1/4 x 14 | 1/4 inch | 14 | 5/16 | 50 | 36 | 45 |
| 1/4 x 20 | 1/4 inch | 20 | 1/4 | 45 | 36 | 53 |
| 3/8 x 4 | 3/8 inch | 4 | 11/16 | 50 | 36 | 48 |
| 3/8 x 6 | 3/8 inch | 6 | 1/2 | 50 | 36 | 48 |
| 3/8 x 8 | 3/8 inch | 8 | 5/16 | 50 | 36 | 48 |
| 3/8 x 10 | 3/8 inch | 10 | 3/8 | 50 | 36 | 48 |
| 3/8 x 12 | 3/8 inch | 12 | 5/16 | 45 | 36 | 48 |
| 3/8 x 16 | 3/8 inch | 16 | 5/16 | 45 | 36 | 48 |
| 3/8 x 20 | 3/8 inch | 20 | 5/16 | 36 | 36 | 48 |
| 1/2 x 4 | 1/2 inch | 4 | 7/16 | 50 | 36 | 48 |
| 1/2 x 6 | 1/2 inch | 6 | 1/2 | 50 | 36 | 48 |
| 1/2 x 8 | 1/2 inch | 8 | 13/32 | 40 | 36 | 48 |
| 1/2 x 10 | 1/2 inch | 10 | 13/32 | 40 | 36 | 48 |
| 1/2 x 12 | 1/2 inch | 12 | 13/32 | 40 | 36 | 48 |

NOTE: Requirements shall be not less than the figures shown except for the rake angle requirements which shall be not more than the figure shown.



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TABLE II. Type II - gap frame, sizes and capacities.

| Machine size | Workpiece thickness | Workpiece width (feet) | Rake angle (inches per foot) | Shear speed (Strokes per minute - full length at rated rake) | Back gauge capacity (inches) | Gap minimum (inches) |
|---|---|---|---|---|---|---|
| 3/16 x 4 | 3/16 inch | 4 | 1/4 | 60 | 48 | 18 |
| 3/16 x 6 | 3/16 inch | 6 | 1/4 | 60 | 48 | 18 |
| 3/16 x 8 | 3/16 inch | 8 | 1/4 | 60 | 48 | 18 |
| 3/16 x 10 | 3/16 inch | 10 | 1/4 | 60 | 48 | 18 |
| 3/16 x 12 | 3/16 inch | 12 | 1/4 | 60 | 48 | 18 |
| 3/16 x 14 | 3/16 inch | 14 | 1/4 | 60 | 48 | 18 |
| 1/4 x 4 | 1/4 inch | 4 | 1/4 | 60 | 36 | 18 |
| 1/4 x 6 | 1/4 inch | 6 | 1/4 | 60 | 36 | 18 |
| 1/4 x 8 | 1/4 inch | 8 | 1/4 | 60 | 36 | 18 |
| 1/4 x 10 | 1/4 inch | 10 | 1/4 | 60 | 36 | 18 |
| 1/4 x 12 | 1/4 inch | 12 | 1/4 | 60 | 36 | 18 |
| 1/4 x 14 | 1/4 inch | 14 | 1/4 | 60 | 36 | 18 |
| 1/4 x 16 | 1/4 inch | 16 | 1/4 | 60 | 36 | 18 |
| 3/8 x 4 | 3/8 inch | 4 | 5/16 | 50 | 36 | 18 |
| 3/8 x 6 | 3/8 inch | 6 | 5/16 | 50 | 36 | 18 |
| 3/8 x 8 | 3/8 inch | 8 | 5/16 | 55 | 36 | 18 |
| 3/8 x 10 | 3/8 inch | 10 | 5/16 | 55 | 36 | 18 |
| 3/8 x 12 | 3/8 inch | 12 | 5/16 | 55 | 36 | 18 |
| 3/8 x 14 | 3/8 inch | 14 | 5/16 | 55 | 36 | 18 |
| 3/8 x 16 | 3/8 inch | 16 | 5/16 | 55 | 48 | 18 |
| 1/2 x 4 | 1/2 inch | 4 | 1/8 | 40 | 36 | 18 |
| 1/2 x 6 | 1/2 inch | 6 | 1/8 | 40 | 36 | 18 |
| 1/2 x 8 | 1/2 inch | 8 | 1/8 | 40 | 36 | 18 |
| 1/2 x 10 | 1/2 inch | 10 | 1/8 | 40 | 36 | 18 |
| 1/2 x 12 | 1/2 inch | 12 | 1/8 | 40 | 36 | 18 |

NOTE: Requirements shall be not less than the figures shown except for the rake angle requirements which shall be not more than the figure shown.



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3.6 Performance. The machine shall be capable of shearing mild steel of not less than 60,000 pounds per square inch (psi) tensile strength in workpiece sizes ranging up to the rated capacity for the thickness and width as specified in table I or II for the type and size machine specified. The rated capacity shall be sheared with the upper shear knife rake angle fixed, if applicable, or set at a rake angle not greater than that specified in table I or II. Shearing shall be accomplished with a shear knife clearance of not more than 15 percent of the workpiece thicknesses up to the maximum capacity of the machine. The sheared edge of the bed cut shall be square with the workpiece surface resting on the table to a tolerance no greater than the knife clearance setting. The workpiece shall be free of visible hold-down cylinder damage.

3.6.1 Shearing accuracy. The machine, while shearing mild steel workpieces of not less than 60,000 PSI tensile strength with the type and size machine specified, shall be capable of shearing 16 gauge material ranging up to the full width capacity of the machine to precision tolerances as follows:

a. Dimensions - The width of the sheared workpiece in relation to the back gauge shall be within $\pm$ 0.002 inch per foot of shear cutting length.

b. Parallelism - The sheared cut shall be parallel to the sheared (trimmed) edge positioned against the back gauge within $\pm$ 0.002 inch per foot of the shear cutting length.

c. Squareness - The sheared cut shall be square with the sheared (trimmed) edge positioned against either side gauge within $\pm$ 0.002 inch per foot of workpiece width.

3.6.2 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.



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3.10.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for

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acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The machine shall be operated at no load for not less than one hour. All controls, motors, adjusting mechanisms, and gauges shall be actuated and observed to determine compliance with the requirements specified herein.



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4.8.3 Performance tests. Performance shall be tested by operating the machine to perform shearing operations to determine conformance with the requirements of 3.6 by shearing mild steel of 60,000 PSI tensile strength. The machine shall shear not less than six workpieces of maximum size and thickness. The instruments and methods used in performing the tests shall be at the option of the contractor, provided they accurately determine the shearing capacity and accuracy of the machine as approved by the contracting officer.

4.8.4 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.5 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station and each end housing area of the machine. Noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

4.8.6 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.8. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.7 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are intended for use in metal shearing and squaring operations that are generally required in all phases of metal fabrication operations. The gap type, type II machines are also intended with optional attachments, for slitting, notching, and angular shearing operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:



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a. Title, number, and date of this specification.

b. Type and size machine required (see 1.2 and table I or II).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Digital readout on shear gauging system for back and/or front gauging, if required (see 3.2.1.2).

g. Specify any special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

j. Noise level, if different (see 3.2.5.2).

k. Environmental protection requirements exceeding EPA/OSHA requirements (see 3.2.8).

l. Lubrication system, if different (see 3.2.9).

m. Specify surface finish requirements, if different (see 3.3.3).

n. Painting, if different (see 3.3.5.).

o. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

p. Specify end housing design, if required (see 3.4.2).

q. Ball transfer arrangement for shears rated 1/4 inch capacity or greater, if required (see 3.4.3).

r. Extra set of shear blades, if required (see 3.4.4.1).

s. Specify gauges, if different (see 3.4.7).

t. If specific type power operated lead screw back gauge is required, state the required type (see 3.4.7.1).

u. Swing up back gauge, if required (see 3.4.7.1).



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v. Side extension squaring gauge, if required (see 3.4.7.3).

w. Light beam shearing gauge, if required (see 3.4.7.4).

x. Adjustable angle gauge, if required (see 3.4.7.5).

y. Slitting gauge, if required (see 3.4.7.6).

z. Voltage requirements, if different (see 3.4.8).

aa. Initial voltage, as specified (see 3.4.8).

bb. Motors, if different (see 3.4.8.1).

cc. Worklight, if required (see 3.4.8.3).

dd. Hour meter, if required (see 3.4.8.4).

ee. Size and capacity, if different (see 3.5).

ff. Specify physical size limitation, if required (see 3.5.1).

gg. Additional performance features, if required (see 3.6.2).

hh. Specify and fully describe optional equipment, if required (see 3.8).

ii. Specify applicable repair parts, if required (see 3.9).

jj. Lubrication plate, if required (see 3.10.1).

kk. Lubrication plate information, if different (see 3.10.1).

ll. Nameplate, if different (see 3.10.2).

mm. Inspection conditions, if different (see 4.3).

nn. First article inspection, if different (see 4.4).

oo. Quality conformance inspection, if different (4.5).

pp. Acceptance test, if different (see 4.6).

qq. Additional performance tests, if required (see 4.8.4).

rr. Noise level acceptance test, if different (see 4.8.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of



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the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| N/A | DI-TMSS-80527 | Commercial Off-The-Shelf (COTS) Manuals | Require all data be in the English Language |
| N/A | DI-TMSS-80528 | Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals | Require all data be in the English Language |

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.



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6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

| MIL-B-80271 | MIL-B-80271A |
|---|---|
| Type I - size 16 ga. x 3 | Deleted |
| | Type II - Gap frame (new type) |
| | Machine sizes Added |
| Size | Size |
| 3/16 x 4 | 3/8 x 4 |
| 3/16 x 6 | 3/8 x 6 |
| 3/16 x 8 | 3/8 x 8 |
| 3/16 x 10 | 3/8 x 10 |
| 3/16 x 12 | 3/8 x 12 |
| 3/16 x 14 | 3/8 x 14 |
| | 3/8 x 16 |
| 1/4 x 4 | 1/2 x 4 |
| 1/4 x 6 | 1/2 x 6 |
| 1/4 x 8 | 1/2 x 8 |
| 1/4 x 10 | 1/2 x 10 |
| 1/4 x 12 | 1/2 x 12 |
| 1/4 x 14 | |
| 1/4 x 16 | |

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

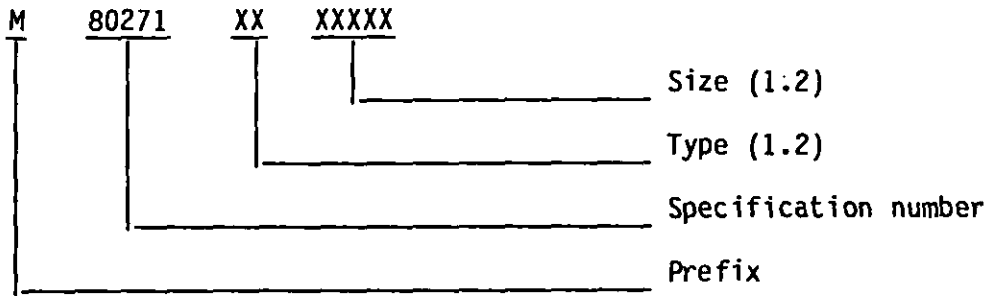
6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.



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6.13 Part or Identifying Number (PIN). The PINs to be used for machines acquired to this specification are created as follows:




6.14 Subject term Key word listing.

Cutting
Metal cutting
Shearing
Slitting
Trimming

Custodians:
Army - AL
Air Force - 99

Review activities:
Air Force - 84
Navy - YD

User activities:
Army - SM, AV
Navy - MC, OS

Preparing Activity:
DLA - IP

Project (3445-0121)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-S-80271A | 92 03 06 |

3. DOCUMENT TITLE
SHEARING MACHINES, METAL SQUARING, MECHANICAL DRIVE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) (1) Commercial | (2) AUTOVON |
|---|---|---|
| Defense Industrial Plant Equipment Center<br>ATTN: DIPEC-SSG | (901)-775-6694 | 683-6694 |
| c. ADDRESS (Include Zip Code)<br>2163 Airways Blvd.<br>Memphis, TN 38114-5213 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

DD Form 1426, OCT 89 Previous editions are obsolete. 198/290

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